Chapter 11 — Learning and Wisdom

Field	Value
Status	Source-aligned rewrite; architecture audit passed
Version	0.2
Scope	Improve HAL through evidence-backed operational learning while protecting constitutional identity and the Owner's values from autonomous modification.
Constitutional basis	Decisions 28, 30, 33–34, 43, 50, 53, 56–58
Owner review	None required

1. Purpose

Improve HAL through evidence-backed operational learning while protecting constitutional identity and the Owner's values from autonomous modification.

2. Authoritative Responsibilities

Component	Authoritative responsibility
Learning Engine	Candidate learning, domains, experiments, promotion and retirement
Learning Ledger	Evidence, versions, promotions, exceptions, adoption and rollback history
Experiment Manager	Control groups, budgets, verification, risk and outcome measurement
Meta-Learning Service	Calibration of learning effectiveness and drift

3. Core State and Records

Object	Required content
Candidate Learning	Proposed lesson, domain, evidence, expected

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	benefit, risk and confidence.
Learning Promotion	Information→Knowledge→Experience→Pattern →Wisdom transition with proof.
Experiment	Hypothesis, variants, allocation, safety envelope, metrics and stopping rules.
Learning Review Tier	Automatic, Notify, Recommend or Constitutional; determines adoption and Owner-authorization requirements.
Wisdom Object	Domain-scoped governed judgment with repeated cross-context evidence and exceptions.

4. Runtime Workflow

1. Record human teaching or self-generated observation as candidate learning, not truth.
2. Assign domain and applicability; prevent cross-domain transfer without evidence.
3. Run controlled, budgeted and reversible experiments where policy permits.
4. Measure outcomes, calibration, exceptions and unintended effects.
5. Adopt low-risk operational improvements according to review tier; preserve rollback.
6. Require Owner Authorization Ceremony for constitutional or protected behavioral change.

5. Interfaces and Contracts

All commands, queries and events are typed, versioned, authenticated and correlated. Commands request mutation from the authoritative owner; queries declare consistency and privacy scope; immutable events record completed facts with identity, causation, authorization, provenance and integrity metadata.

6. Failure and Recovery

- Insufficient evidence: keep the candidate provisional.
- Experiment harms quality or violates guardrail: stop and rollback within the declared envelope.
- Learning drift: quarantine the affected domain and restore the last verified behavior.
- Owner rejects proposal: retain the decision and evidence without covert adoption.

7. Constitutional Guarantees

- HAL may improve indefinitely but not at the expense of constitutional identity.
- Wisdom is earned through repeated evidence and changes slowly.
- Every adopted change has evidence, review tier, outcome and rollback history.

- Learning effort is budgeted and subordinate to serving the Owner.

8. Prohibited Behaviors

- No component may silently broaden authority, reinterpret Owner intent, erase dissent, or treat a derived projection as authoritative state.
- No degraded dependency may silently change policy, identity, trust, authentication, verification or evidence requirements.
- No provider, model, node, Presence, secret, relationship or network location receives constitutional authority by implication.

9. Observability and Verification

The subsystem records correlation, identity, policy/authority context, state versions, decisions, limitations, failures, recovery and outcomes. Verification is risk-scaled and produces durable evidence. Status reporting is derived from governed state rather than model assertion.

10. Source Alignment and Review

This chapter directly implements Decisions 28, 30, 33–34, 43, 50, 53, 56–58. The v0.2 rewrite restores the source-specific objects, workflows and guarantees missing from v0.1. Final disposition is recorded in the separate source-alignment and constitutional-audit reports.